YE Futao

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github.com/Fttt1003/futao-ye-portfolio

Available from early December 2026 | Interested in software development and AI interaction

EDUCATION

The Hong Kong Polytechnic University | Fall 2026 - Present

MSc in Metaverse Technology | Expected graduation: 2027

South China Normal University | September 2022 - September 2026

BSc in Software Engineering | Average: 84.1/100

SELECTED PROJECTS

Movie Application | Java Backend Development

- Contributed movie query, filtering and user-favourites functionality using Spring Boot and MyBatis within a RuoYi-based application.
- The project includes paginated catalogue responses, actor/director relationships and Excel export of enriched movie records.

Road Assistance for Visually Impaired Users | March 2024 - June 2025

- Prepared and augmented segmentation masks from two datasets; trained and evaluated DeepLabv3+ for road-scene segmentation.
- Built an Android location and route-guidance prototype with interfaces reserved for future model integration.

License Plate Recognition | Python and OpenCV Coursework

- Built a pipeline for mainland Chinese and Hong Kong plates using HSV masking, morphology, perspective rectification and projection-based character segmentation.
- Combined image difference, edge similarity and contour matching for template recognition; visualised intermediate outputs for debugging.

INTERNSHIP EXPERIENCE

Nanhua Instrument Co., Ltd. | September - December 2025

IoT Software Development Intern

- Supported device monitoring and data visualisation for smart gas cabinets; integrated EMQX/MQTT communication between devices and cloud services.

Foshan Electric Power Design Institute | January - February 2025

Research and Development Intern

- Supported requirements analysis, first-version design and technical validation of an internal innovation competition platform.

PUBLICATIONS

Ye, F. (2025). Rewriting K-Means Clustering Algorithm Using DataFrames API. ICCSM 2025, 457-460.

Ye, F. (2025). A Survey of Research Advances in Federated Learning for Anomaly Detection. Science and Technology of Engineering, Chemistry and Environmental Protection, 3(5).

SKILLS AND CERTIFICATION

Python, Java, OpenCV, NumPy, scikit-learn, Spark DataFrame API, Spring Boot, MyBatis, EMQX/MQTT.

Huawei Advanced HarmonyOS Developer Certification (October 2024).